AI EXPLAINED SIMPLY

AI Agent Design Template

A ready-to-copy way to design your first AI agent, from the video "AI Agents Explained in 10 Minutes".

The idea in one line

An AI agent is a language model that runs in a loop: **plan, act, observe, check**, with tools that touch the real world and a rule that says when to stop.

Workflow or agent? (Anthropic, Dec 2024)

- **Workflows** are systems where LLMs and tools are orchestrated through predefined code paths.
- **Agents** are systems where LLMs dynamically direct their own processes and tool usage.
- Advice from the same guide: start with the simplest solution, and add an agent only when a fixed workflow is not enough.

The loop

1. **Plan**: read the goal, pick the next step (only the next one).
2. **Act**: call a tool (search, read a file, run code, send an email).
3. **Observe**: read the real result, including errors.
4. **Check**: am I done? If yes, answer. If not, plan again.

Three questions before you build

1. Can the steps be written down in advance? Then use a workflow, not an agent.
2. Does the task need the real world (search, files, code)? Then it needs tools.

3. Can a wrong action be undone? If not, add a human approval step.

Copy-paste design template

```
GOAL: <one sentence>
TOOLS: <tool 1>, <tool 2>, <tool 3>
STEP LIMIT: 10
STOP WHEN: the goal is met, or the step limit is reached
APPROVAL NEEDED FOR: anything that cannot be undone (sending, deleting, paying)
```

Starter tasks

- Sort your inbox into folders.
- Summarize five web pages into one page.
- Check a spreadsheet for duplicates.

Four ways agents fail

- They loop forever: set a step limit.
- Errors compound: one wrong observation can poison later steps.
- They act in the real world: give the smallest set of permissions, keep a human approval step.
- They cost more: every loop is another model call.

Sources

- Anthropic, "Building Effective Agents", 19 Dec 2024: https://www.anthropic.com/engineering/building-effective-agents
- Yao et al., "ReAct: Synergizing Reasoning and Acting in Language Models", arXiv 2210.03629: https://arxiv.org/abs/2210.03629

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